

Loan DataSet

Column Name Description

Loan_ID Unique loan identifier

Gender Male / Female

Married Applicant marital status

Dependents Number of dependents

Education Graduate / Not Graduate

Self_Employed Self employment status

ApplicantIncome Applicant monthly income

CoapplicantIncome Co-applicant income

LoanAmount Loan amount requested

Loan_Amount_Term Term of loan (in months)

Credit_History 1 = good credit, 0 = bad credit

Property_Area Urban / Semiurban / Rural

Loan_Status Y = Approved, N = Rejected

Data Analysis Using Pyspark (15 Marks)

Create Spark session and load dataset

Display schema of dataset

Show first 10 rows

Count total rows and columns in dataset s

Check missing values in each column

Find count and percentage of each class in Loan_Status (imbalance check)

Find number of applicants grouped by Gender and Loan_Status -

Compute average ApplicantIncome for each Education category

Find median LoanAmount for each Property_Area

Identify outliers in ApplicantIncome using IQR method

Find correlation between ApplicantIncome, LoanAmount, and Credit_History

Identify which combination of (Education + Property_Area) has the highest loan approval rate.

Find whether co-applicant income significantly impacts loan approval (compare averages for Y vs N).

Analyze if higher loan amount leads to lower approval probability (bin LoanAmount and compare trends).

Determine the rejection rate for applicants with Credit_History = 0 across all Property_Areas.

Find top 3 patterns of applicants most likely to get loan approval (based on grouping attributes).

MapReduce Section (2 Questions=20 Marks)

Q1 Find the average LoanAmount for each Property_Area using MapReduce.

Q2 Compute the count of approved and rejected loans using MapReduce.

Q3 Find the maximum ApplicantIncome for each Education category using MapReduce.

Q4 Find the number of applicants grouped by Gender and Loan_Status using MapReduce.

Q5 Compute the average CoapplicantIncome for each Loan_Status category (Y/N) using MapReduce.

Airflow Section (5 Marks)

Task:

Write a Spark job that:

Loads loan_prediction_dataset.csv Filters applicants where: Credit_History = 1 AND Loan_Status = Y Saves output as a new file

DAG Requirements:

Create an Airflow DAG that:

Runs the Spark job Executes every 2 hours Stores output in a structured folder (HDFS/local) Logs execution status